

Lec 23

$$A = \{X \in B\}$$

$$F_{X|A}(x) = \frac{P(\{X \leq x\} \cap \{X \in B\})}{P(\{X \in B\})}$$

$$= \int_{\substack{u \leq x \\ u \in B}} f_X(u) du$$

$$= \frac{\int_{-\infty}^x f_X(u) 1_{\{u \in B\}} du}{P(X \in B)}$$

$$\Rightarrow f_{X|A}(x) = \frac{f_X(x) 1_{\{x \in B\}}}{P(X \in B)} = \begin{cases} \frac{f_X(x)}{P(X \in B)} & x \in B \\ 0 & \text{otherwise} \end{cases}$$

Total Probability theorem.

$A_1, A_2, \dots, A_n$ that partition Ω

$$P(A) = \sum_{i=1}^n P(A \cap A_i) = \sum_{i=1}^n P(A|A_i)P(A_i).$$

$$P_X(x) = \sum_{i=1}^n P_{X|A_i}(x) P(A_i).$$

$$f_X(x) = \sum_{i=1}^n f_{X|A_i}(x) P(A_i).$$

$$F_X(x) = P(\{X \leq x\})$$

$$A = \{X \leq x\} = \sum_{i=1}^n P(\{X \leq x\} | A_i) P(A_i) = \sum_{i=1}^n F_{X|A_i}(x) P(A_i)$$

Total Probability theorem

$A_1, A_2, \dots, A_n$ that partition Ω

$$P(A) = \sum_{i=1}^n P(A \cap A_i) = \sum_{i=1}^n P(A|A_i)P(A_i)$$

$$P_X(x) = \sum_{i=1}^n P_{X|A_i}(x) P(A_i)$$

$$f_X(x) = \sum_{i=1}^n f_{X|A_i}(x) P(A_i)$$

$$F_X(x) = P(\{X \leq x\})$$

$$A = \{X \leq x\} = \sum_{i=1}^n P(\{X \leq x\} | A_i) P(A_i) = \sum_{i=1}^n F_{X|A_i}(x) P(A_i)$$

Suppose $f_{X|A_i}$ is pdf of $X|A_i$

$$\text{ie, } F_{X|A_i}(x) = \int_{-\infty}^x f_{X|A_i}(u) du$$

$$\text{RHS} = \sum_{i=1}^n \left[\int_{-\infty}^x f_{X|A_i}(u) du \right] P(A_i)$$

$$= \int_{-\infty}^x \underbrace{\left[\sum_{i=1}^n f_{X|A_i}(u) P(A_i) \right]}_{f_X(u)} du$$

$$= F_X(x)$$

$$\Rightarrow \boxed{f_X(x) = \sum_{i=1}^n f_{X|A_i}(x) P(A_i)}$$

$$f_{X|A_i}(x) P(A_i)$$

Example:

Hotel $\rightarrow$ Main gate bus arrives every 15 mins starting at 6am. You walk to the boarding point every morning between 7.10-7.30 with time interval being uniform in that interval.

What is the pdf of the waiting time?

X : RV indicating your arrival time at boarding point.

$X \sim \text{Unif. form } [7.10, 7.30]$

$$f_X(x) = \frac{1}{20} \quad x \in [7.10, 7.30] \text{ (mins units)}$$

$$A_1 = \{X \leq 7.15\}$$

$$A_1^c = \{X > 7.15\}$$

Waiting times

Y : waiting time incurred

$$Y|A_1 \quad Y = 7.15 - X$$

$$Y|A_2 \quad Y = 7.30 - X$$

To find
 $f_Y(y)$

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$$f_Y(y) = f_{Y/A_1}(y) P(A_1) + f_{Y/A_1^c}(y) P(A_1^c)$$

$$A_1 = \{X \leq 7.15\}.$$

$$X \sim \text{Uniform}[7.10, 7.30] \quad f_X(x) = \frac{1}{20}.$$

$$P(A_1) = \int_{7.10}^{7.15} f_X(x) dx.$$

$$= \frac{5}{20} = \frac{1}{4}.$$

$$P(A_1^c) = 3/4.$$

$$f_{X|A_1}(x) = \begin{cases} \frac{f_X(x)}{P(A_1)} & x \leq 7.15 \\ 0 & \text{otherwise} \end{cases} = \begin{cases} 1/5 & 7.10 \leq x \leq 7.15 \\ 0 & \text{otherwise} \end{cases}$$

$$f_{X|A_2}(x) = \begin{cases} \frac{f_X(x)}{P(A_2)} & x > 7.15 \\ 0 & \text{otherwise} \end{cases} = \begin{cases} 1/15 & 7.15 \leq x \leq 7.30 \\ 0 & \text{otherwise} \end{cases}$$

$$X|A_1 \sim \text{Uniform}[7.10, 7.15]$$

$$X|A_2 \sim \text{Uniform}[7.15, 7.30]$$

$$\begin{aligned} U &\sim \text{Uniform}[a, b] \\ &\equiv U + m \\ &\sim \text{Uniform}[a+m, b+m] \end{aligned}$$

$$Y = 7.15 - X \quad | A_1, \quad Y = 7.30 - X \quad | A_2$$

$$Y|A_1 \sim \text{Uniform}[0, 5]$$

$$Y|A_2 \sim \text{Uniform}[0, 15]$$

$$f_{Y|A_1}(y) = \frac{1}{5} \quad y \in \{0, 5\}$$

$$f_{Y|A_2}(y) = \frac{1}{15} \quad y \in \{0, 15\}$$

$$f_Y(y) = P(A_1) f_{Y|A_1}(y) + P(A_1^c) f_{Y|A_1^c}(y)$$

$$= \begin{cases} \frac{1}{4} \times \frac{1}{5} + \frac{3}{4} \times \frac{1}{15} & y \in [0, 5] \\ \frac{3}{4} \times \frac{1}{15} & y \in (5, 15] \end{cases}$$

$$= \begin{cases} \frac{1}{10} & y \in [0, 5] \\ \frac{1}{20} & y \in (5, 15] \end{cases}$$

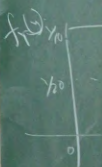
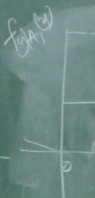
$$E[Y|A_1] = \frac{5}{2}$$

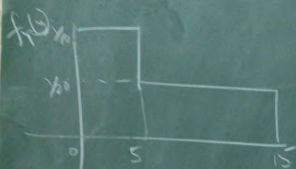
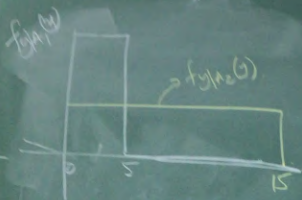
$$\Rightarrow E[Y] = P(A_1) \times \frac{5}{2}$$

$$+ P(A_1^c) \times 15$$

$$E[Y|A_1^c] = \frac{15}{2}$$

$$= \frac{1}{4} \times \frac{5}{2} + \frac{3}{4} \times \frac{15}{2} = \frac{20}{8} = 6.25$$





$$\int_0^{15} f(x) dx = 5 \times \frac{1}{10} + 10 \times \frac{1}{20} = 1$$

$$E[X] = \int_0^{\infty} y f(y) dy$$

$$= \int_0^5 y \times \frac{1}{10} dy + \int_5^{15} y \times \frac{1}{20} dy$$

6.25

Expectation conditioned over an event

For x discrete

$$E[X|A] = \sum_{x \in A} x P_{X|A}(x)$$

$$E[X|A] = \int x f_{X|A}(x) dx$$

$A_1, A_2, \dots$ - An partition Ω

$$E[X] = \sum_{i=1}^n E[X|A_i] P(A_i)$$

$$E[X] = \int_{-\infty}^{\infty} x f_X(x) dx$$

$$= \int_{-\infty}^{\infty} x \left[\sum_{i=1}^n f_{X|A_i}(x) P(A_i) \right] dx$$